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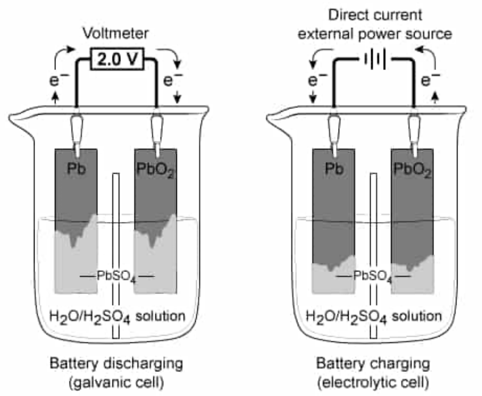


Figure 1 Electron flow in a lead storage battery when discharging and charging

The half-reactions for the anode and cathode of a discharging lead storage battery in 4 M sulfuric acid (H_2SO_4) are shown in Reactions 1 and 2, respectively:

$$\mathrm{Pb}(s) + \mathrm{HSO_4^-}(aq) \rightarrow \mathrm{PbSO_4}(s) + \mathrm{H^+}(aq) + 2\,\mathrm{e^-}$$

Reaction 1

$$\mathrm{PbO_2}(s) + \mathrm{HSO_4^-}(aq) + 3\,\mathrm{H^+}(aq) + 2\,\mathrm{e^-} \rightarrow \mathrm{PbSO_4}(s) + 2\,\mathrm{H_2O}(l)$$

Reaction 2

NiCd batteries have a greater energy-to-weight ratio than lead storage batteries but cannot hold as much charge. Each NiCd cell has an E_{cell} of approximately 1.3 V, and a single-cell battery weighing 120 g can provide 7.2 W·h of energy when discharged.

The half-reactions at the anode and cathode for a discharging NiCd battery in KOH(*aq*) are shown in Reactions 3 and 4, respectively:

$$\mathrm{Cd}(s) + 2\,\mathrm{OH^-}(aq) \rightarrow \mathrm{Cd(OH)_2}(s) + 2\,e^-$$

Reaction 3

$$2\,\mathrm{NiO_2}(s) + 2\,\mathrm{H_2O}(l) + 2\,\mathrm{e^-} \rightarrow 2\,\mathrm{Ni(OH)_2}(s) + 2\,\mathrm{OH^-}(aq)$$

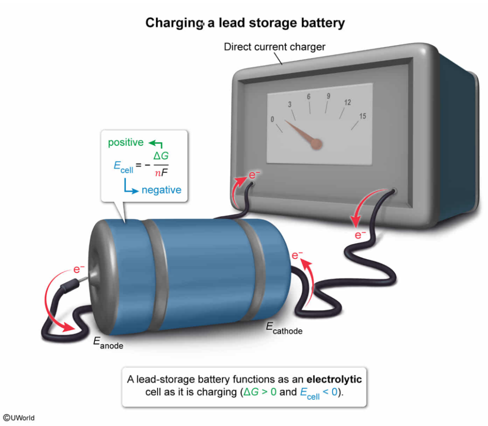
Reaction 4

If a single-cell lead storage battery in an AED is charging, it is operating as a(n):

- A. galvanic cell with a positive E_{cell}. [13%]
- ✔ B. electrolytic cell with a negative E_{cell}. [80%]
- C. concentration cell with a negative E_{cell}. [3%]
- D. fuel cell with a positive E_{cell}. [3%]

Correct
79% answered correctly

Explanation:



In an electrochemical cell, a redox reaction converts chemical energy to electrical energy (ie, a galvanic cell) or vice versa (ie, an electrolytic cell). The Gibbs free energy ΔG produced from these energy conversions represents the maximum amount of work that can be performed during the reaction. The amount of work required to move one unit of charge through an electrochemical cell is the **electromotive force** (EMF) and is inferred from the **cell potential E_{cell}** (ie, the voltage difference between the anode and cathode before current begins to flow). As such, ΔG and E_{cell} are directly related by

$$\Delta G = -\mathrm{nF}E_{\mathrm{cell}}$$

or equivalently,

$$E_{\mathrm{cell}} = -\frac{\Delta G}{\mathrm{nF}}$$

where n is the number of electrons transferred per reaction and F is Faraday's constant.

In a galvanic cell, a spontaneous reaction occurs (ie, **$\Delta G < 0$**; **$E_{cell} > 0$**). In an electrolytic cell, **an external potential source** is used to drive a nonspontaneous reaction (ie, **$\Delta G > 0$**; **$E_{cell} < 0$**). Consequently, an electrolytic cell can be viewed as a galvanic cell operating in reverse.

During the charging phase of a lead storage battery, an external potential is applied to force the redox reaction to proceed in a nonspontaneous direction ($\Delta G > 0$). Therefore, the battery functions as an electrolytic cell with a negative E_{cell}.

(Choice A) A galvanic cell converts chemical energy to electrical energy via a spontaneous redox reaction ($\Delta G < 0$; $E_{cell} > 0$).

(Choice C) A concentration cell is a type of galvanic cell in which electrons flow from a lower concentration half-cell into a higher concentration half-cell, thus generating a potential. Because the concentration difference is the driving force behind the electron transfer, E_{cell} is positive.

(Choice D) A fuel cell is a type of galvanic cell in which the reactants are continuously supplied at the anode and cathode as the products are continuously removed from the system. Although outside material is continuously supplied to a fuel cell, its reaction is still spontaneous and E_{cell} is positive.

Educational objective:
When a battery is charging, it functions as an electrolytic cell because an external potential drives the oxidation-reduction reaction to proceed in a nonspontaneous direction (ie, $\Delta G > 0$; $E_{cell} < 0$).

Time spent:0 sec
Subject:
General Chemistry

QID:401279
System:
Solutions and Electrochemistry

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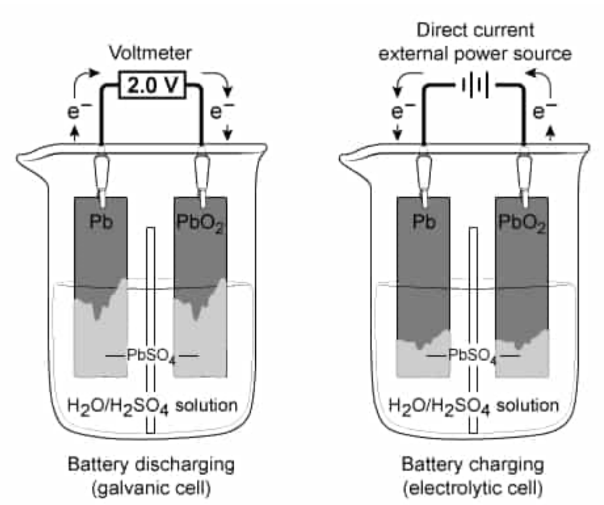


Figure 1 Electron flow in a lead storage battery when discharging and charging

The half-reactions for the anode and cathode of a discharging lead storage battery in 4 M sulfuric acid (H_2SO_4) are shown in Reactions 1 and 2, respectively:

$$\mathrm{Pb}(s) + \mathrm{HSO_4^-}(aq) \rightarrow \mathrm{PbSO_4}(s) + \mathrm{H^+}(aq) + 2\,\mathrm{e^-}$$

Reaction 1

$$\mathrm{PbO_2}(s) + \mathrm{HSO_4^-}(aq) + 3\,\mathrm{H^+}(aq) + 2\,\mathrm{e^-} \rightarrow \mathrm{PbSO_4}(s) + 2\,\mathrm{H_2O}(l)$$

Reaction 2

NiCd batteries have a greater energy-to-weight ratio than lead storage batteries but cannot hold as much charge. Each NiCd cell has an E_{cell} of approximately 1.3 V, and a single-cell battery weighing 120 g can provide 7.2 W·h of energy when discharged.

The half-reactions at the anode and cathode for a discharging NiCd battery in KOH(*aq*) are shown in Reactions 3 and 4, respectively:

$$\mathrm{Cd}(s) + 2\,\mathrm{OH}^-(\mathrm{aq}) \rightarrow \mathrm{Cd(OH)_2}(s) + 2\,e^-$$

Reaction 3

$$2\,\mathrm{NiO_2}(\mathrm{s}) + 2\,\mathrm{H_2O}(l) + 2\,\mathrm{e}^- \rightarrow 2\,\mathrm{Ni(OH)_2}(s) + 2\,\mathrm{OH}^-(\mathrm{aq})$$

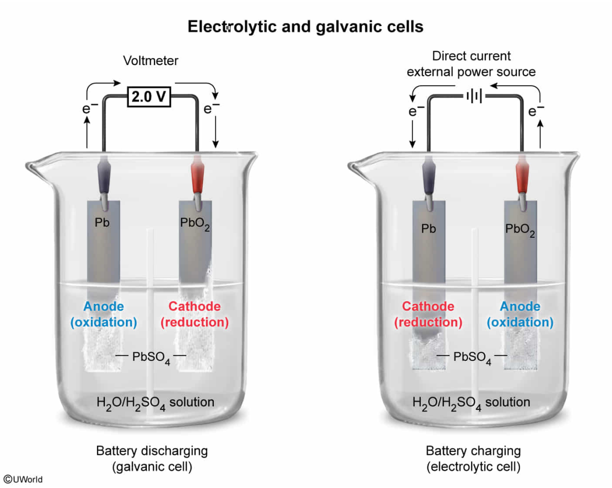
Reaction 4

Which statement correctly describes the Pb electrode when the lead storage battery is charging, as shown in Figure 1?

- A. Pb is the anode and $PbSO_4$ is oxidized at its surface. [21%]
- B. Pb is the anode and H_2SO_4 is reduced at its surface. [11%]
- C. Pb is the cathode and $PbSO_4$ is reduced at its surface. [50%]
- D. Pb is the cathode and H_2SO_4 is oxidized at its surface. [15%]

Correct
51% answered correctly

Explanation:



When a battery is **discharging**, a spontaneous redox reaction occurs, producing an electric current (ie, the battery operates as a galvanic cell). Conversely, when a battery is **charging**, an electric current supplied from an external source is used to drive a nonspontaneous redox reaction (ie, the battery operates as an electrolytic cell).

Although electron flow is in the opposite direction when a battery is discharging versus when it is charging, electrons always flow from the anode (ie, the electrode where **oxidation** occurs) to the **cathode** (ie, the electrode where **reduction** occurs). In effect, the electrodes switch roles when the cell reverses from discharging to charging.

As the lead-acid battery described in the passage is discharging, $PbSO_4$(*s*) forms and accumulates on the surface of both the Pb and PbO_2 electrodes (Reactions 1 and 2). According to Figure 1, when the lead-acid battery is charging (ie, the reverse of discharging), electrons flow from the PbO_2 electrode toward the Pb electrode. Therefore, the Pb electrode is the cathode and because reduction occurs at the cathode, $PbSO_4$(*s*) is reduced (ie, the reverse of Reaction 1) as the battery is charging.

(Choice A) Pb(*s*) acts as the anode (ie, undergoes oxidation) when the battery is *discharging* (ie, electrons flow from Pb to PbO_2) but when the battery is *charging* (ie, electrons flow from PbO_2 to Pb), Pb(*s*) acts as the cathode (ie, $PbSO_4$(*s*) is *reduced* at the cathode surface).

(Choices B and D) In a lead-acid battery, H_2SO_4 is the electrolyte. Although the electrolyte serves as a charge carrier, it is neither oxidized nor reduced by the reaction.

Educational objective:
When a battery is discharging, it behaves as a galvanic cell; when it is charging, it behaves as an electrolytic cell. In either case, oxidation occurs at the anode and reduction occurs at the cathode, and electrons flow from the anode to the cathode.

Time spent:0 sec
Subject:
General Chemistry

QID:401280
System:
Solutions and Electrochemistry

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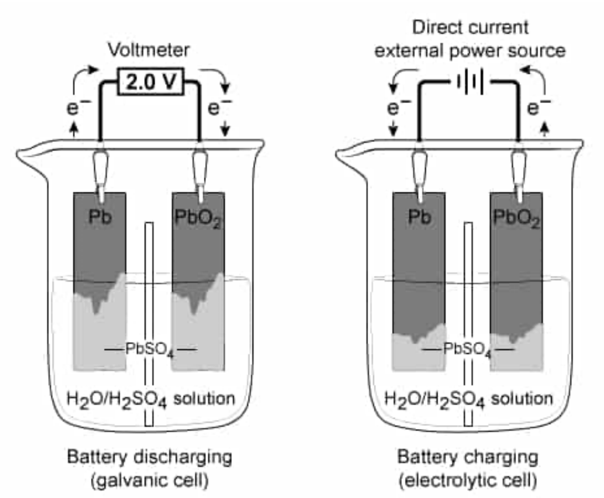


Figure 1 Electron flow in a lead storage battery when discharging and charging

The half-reactions for the anode and cathode of a discharging lead storage battery in 4 M sulfuric acid (H_2SO_4) are shown in Reactions 1 and 2, respectively:

$$\mathrm{Pb}(s) + \mathrm{HSO_4^-}(aq) \rightarrow \mathrm{PbSO_4}(s) + \mathrm{H^+}(aq) + 2\,\mathrm{e^-}$$

Reaction 1

$$\mathrm{PbO_2}(s) + \mathrm{HSO_4^-}(aq) + 3\,\mathrm{H^+}(aq) + 2\,\mathrm{e^-} \rightarrow \mathrm{PbSO_4}(s) + 2\,\mathrm{H_2O}(l)$$

Reaction 2

NiCd batteries have a greater energy-to-weight ratio than lead storage batteries but cannot hold as much charge. Each NiCd cell has an E_{cell} of approximately 1.3 V, and a single-cell battery weighing 120 g can provide 7.2 W·h of energy when discharged.

The half-reactions at the anode and cathode for a discharging NiCd battery in

KOH(*aq*) are shown in Reactions 3 and 4, respectively:

$$\mathrm{Cd}(s) + 2\ \mathrm{OH}^-(\mathrm{aq}) \rightarrow \mathrm{Cd(OH)}_2(s) + 2\ e^-$$

Reaction 3

$$2\ \mathrm{NiO_2(s)} + 2\ \mathrm{H_2O}(l) + 2\ \mathrm{e}^- \rightarrow 2\ \mathrm{Ni(OH)}_2(s) + 2\ \mathrm{OH}^-(\mathrm{aq})$$

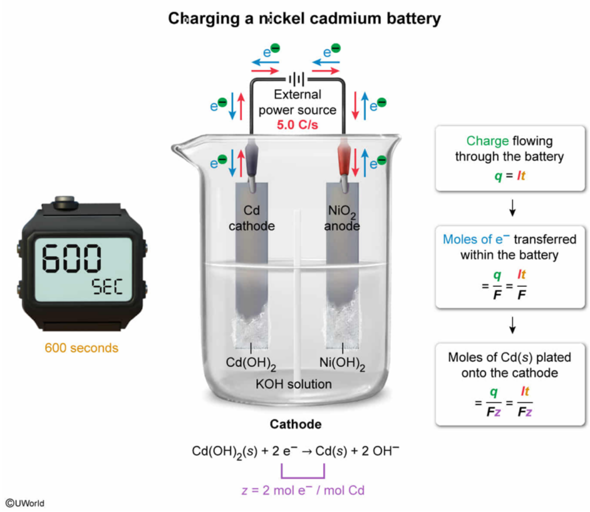
Reaction 4

Assume that a NiCd battery charges for a time (*t*) of 600 seconds with an applied current (*I*) of 5.0 A. Which expression could be used to calculate the number of moles of Cd deposited on the cathode surface by electrolytic plating while charging? (Note: *z* = 2 mol e⁻ / mol Cd, and *F* = 9.6 × 10^4 C / mol e⁻.)

- A. $\frac{\mathrm{Fz}}{\mathrm{It}}$ [18%]
- B. $\frac{\mathrm{FI}}{\mathrm{zt}}$ [17%]
- C. $\frac{I}{\mathrm{Fzt}}$ [10%]
- ✔ D. $\frac{\mathrm{It}}{\mathrm{Fz}}$ [54%]

Correct
54% answered correctly

Explanation:



In a rechargeable battery, the spontaneous redox reaction occurring during discharge is reversed by applying an electric current from an external source. This nonspontaneous process, known as **electrolysis**, electrochemically decomposes a compound by oxidizing electron-rich species at the anode of the electrolytic cell while depositing electron-poor species onto the cathode after reducing them. The electrolytic deposition of solids onto a metal surface is known as **electroplating** and involves the reduction of a **metal ion** to a **metal solid** at the cathode surface.

As a NiCd battery is charging, Cd^{2+} ions in $Cd(OH)_2(s)$ are reduced to Cd(*s*) (ie, Reaction 3 in reverse) and plated onto the cathode surface:

$$\mathrm{Cd(OH)_2}(s) + 2\,\mathrm{e}^- \rightarrow \mathrm{Cd}(s) + 2\,\mathrm{OH}^-(\mathrm{aq})$$

Electric current is measured in amperes, where 1 ampere (A) equals 1 coulomb (C) of electric charge flowing through the battery per second. The question states that a current of 5.0 A (ie, 5.0 C/s) is applied to the NiCd battery for 600 s. The total electric charge *q* (in units of C) flowing through the battery during the given interval can be determined by multiplying the current *I* by the time *t*:

$$q = I\left(\frac{\mathrm{C}}{\cancel{\mathrm{s}}}\right) \times \mathrm{t}(\cancel{\mathrm{s}}) = \mathrm{It}(\mathrm{C})$$

Faraday's constant *F* represents the amount of electric charge carried by 1 mole of electrons. Dividing *q* by *F* yields the total moles of electrons transferred within the battery:

$$\frac{q}{F} = \frac{\mathrm{It}(\cancel{\mathrm{C}})}{F\left(\frac{\cancel{\mathrm{C}}}{\mathrm{mol\ e}^-}\right)} = \frac{\mathrm{It}}{F}(\mathrm{mol\ e}^-)$$

Based on Reaction 3 (and as noted in the question), 2 moles of electrons are required to reduce 1 mole of Cd(*s*) (ie, $z = 2$ mol e^- / mol Cd). Therefore, the total moles of Cd(*s*) deposited on the cathode surface by electrolytic plating can be expressed as

$$\frac{\mathrm{It}}{F}(\cancel{\mathrm{mol\ e}^-}) \times \frac{1}{z\left(\frac{\cancel{\mathrm{mol\ e}^-}}{\mathrm{mol\ Cd}}\right)} = \frac{\mathrm{It}}{\mathrm{Fz}}(\mathrm{mol\ Cd})$$

Although a numerical value is not required for this question, substituting the values into the above expression yields:

$$\frac{\mathrm{It}}{\mathrm{Fz}} = \frac{\left(5.0\frac{\cancel{\mathrm{C}}}{\cancel{\mathrm{s}}}\right)(600\ \cancel{\mathrm{s}})}{\left(\frac{9.6\times10^4\ \cancel{\mathrm{C}}}{\cancel{\mathrm{mol\ e}^-}}\right)\left(\frac{2\ \cancel{\mathrm{mol\ e}^-}}{\mathrm{mol\ Cd}}\right)} = 0.015\ \mathrm{mol\ Cd}$$

(Choice A) This equation is the reciprocal of the correct equation and gives the answer in terms of *inverse moles* of Cd(*s*).

(Choices B and C) All the units should cancel in the expression to give moles of Cd(*s*). In these expressions, the units do not cancel to yield the desired result.

Educational objective:

In an electrolytic cell, the deposition of solids by the reduction of a metal ion onto the cathode surface is known as electroplating. Using the number of electrons (*z*) transferred per reaction and Faraday's constant (*F*), the number of moles (*n*) of metal plated onto a surface using current (*I*) over time (*t*) is given by the equation, $n = \frac{\mathrm{It}}{\mathrm{Fz}}$.

Time spent: 0 sec
Subject: General Chemistry

QID: 401281
System: Solutions and Electrochemistry

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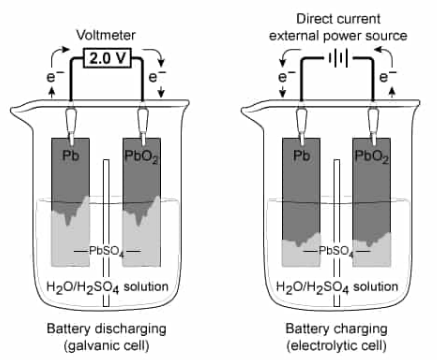


Figure 1 Electron flow in a lead storage battery when discharging and charging

The half-reactions for the anode and cathode of a discharging lead storage battery in 4 M sulfuric acid (H_2SO_4) are shown in Reactions 1 and 2, respectively:

$$\mathrm{Pb}(s) + \mathrm{HSO_4^-}(\mathrm{aq}) \rightarrow \mathrm{PbSO_4}(s) + \mathrm{H^+}(\mathrm{aq}) + 2\,\mathrm{e^-}$$

Reaction 1

$$\mathrm{PbO_2}(s) + \mathrm{HSO_4^-}(\mathrm{aq}) + 3\,\mathrm{H^+}(\mathrm{aq}) + 2\,\mathrm{e^-} \rightarrow \mathrm{PbSO_4}(s) + 2\,\mathrm{H_2O}(l)$$

Reaction 2

NiCd batteries have a greater energy-to-weight ratio than lead storage batteries but cannot hold as much charge. Each NiCd cell has an E_{cell} of approximately 1.3 V, and a single-cell battery weighing 120 g can provide 7.2 W·h of energy when discharged.

The half-reactions at the anode and cathode for a discharging NiCd battery in KOH(*aq*) are shown in Reactions 3 and 4, respectively:

$$\mathrm{Cd}(s) + 2\,\mathrm{OH^-}(\mathrm{aq}) \rightarrow \mathrm{Cd(OH)_2}(s) + 2\,e^-$$

Reaction 3

$$2\,\mathrm{NiO_2}(\mathrm{s}) + 2\,\mathrm{H_2O}(l) + 2\,\mathrm{e^-} \rightarrow 2\,\mathrm{Ni(OH)_2}(s) + 2\,\mathrm{OH^-}(\mathrm{aq})$$

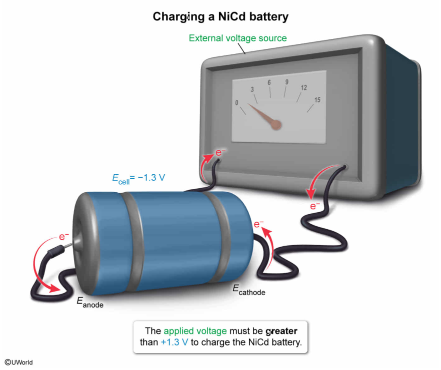
Reaction 4

The passage states that a typical single-cell NiCd battery has an average cell potential of 1.3 V when discharged. To charge a single-cell NiCd battery, the applied external potential must be:

- ✔ A. greater than 1.3 V. [59%]
- B. equal to 1.3 V. [12%]
- C. less than −1.3 V. [10%]
- D. equal to −1.3 V. [17%]

Correct
58% answered correctly

Explanation:



In an electrochemical cell, a redox reaction converts chemical energy to electrical energy (ie, a discharging battery) or vice versa (ie, a charging battery). The Gibbs free energy ΔG produced from these energy conversions represents the maximum amount of work that can be performed during the reaction. The amount of work required to move one unit of charge through the electrochemical cell is the **electromotive force** (EMF) and is inferred from the **cell potential E_{cell}** (ie, the voltage difference between the anode and cathode before current begins to flow). As such, ΔG and E_{cell} are directly related by

$$\Delta G = -\mathrm{nF}E_{\mathrm{cell}}$$

or equivalently,

$$E_{\mathrm{cell}} = -\frac{\Delta G}{\mathrm{nF}}$$

where n is the number of electrons transferred per reaction and F is Faraday's constant.

When a battery is discharging, a spontaneous reaction occurs (ie, $\boldsymbol{\Delta G < 0}$; $\boldsymbol{E_{cell} > 0}$). Conversely, when a battery is charging, an **external voltage source** is used to drive the spontaneous discharge reaction in the nonspontaneous, reverse direction (ie, $\boldsymbol{\Delta G > 0}$; $\boldsymbol{E_{cell} < 0}$).

To reverse the discharge reaction, the net ΔG of the combined system (ie, the charging battery and external voltage source) must be **negative**:

$$\Delta G_{\text{net}} = \Delta G_{\text{external source}} + \Delta G_{\text{charging battery}} < 0$$

As such, the net cell potential must be **positive**:

$$E_{\text{net}} = E_{\text{external source}} + E_{\text{charging battery}} > 0$$

Consequently, an external voltage that is **greater** than the voltage generated by discharging the battery is required to charge it:

$$E_{\text{external source}} + (-1.3\ \text{V}) > 0$$

$$E_{\text{external source}} > 1.3\ \text{V}$$

The question states that a single-cell NiCd battery generates an average cell potential of +1.3 V when discharged. Therefore, charging the battery would require more than +1.3 V of applied potential to drive the reaction in the nonspontaneous direction.

(Choice B) For the nonspontaneous reaction to occur, the applied potential must be *greater than* (not equal to) the potential that the oxidation-reduction reaction spontaneously produces when discharging.

(Choices C and D) Although the cell potential for a nonspontaneous reaction is negative, the *applied potential* must be *positive* to overcome the negative potential of the reverse reaction.

Educational objective:
When charging a battery, an external positive electric potential must be applied to force the spontaneous ($\Delta G < 0$) oxidation-reduction reaction in the nonspontaneous ($\Delta G > 0$) direction. The net ΔG for the system must be negative to charge the battery.

Time spent:0 sec	**QID**:401282
Subject:	**System:**
General Chemistry	Solutions and Electrochemistry

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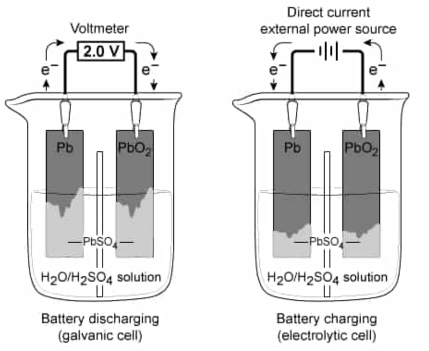

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Reaction 2

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$$\mathrm{Cd}(s) + 2\,\mathrm{OH^-}(aq) \rightarrow \mathrm{Cd(OH)_2}(s) + 2\,e^-$$

Reaction 3

$$2\,\mathrm{NiO_2}(s) + 2\,\mathrm{H_2O}(l) + 2\,\mathrm{e^-} \rightarrow 2\,\mathrm{Ni(OH)_2}(s) + 2\,\mathrm{OH^-}(aq)$$

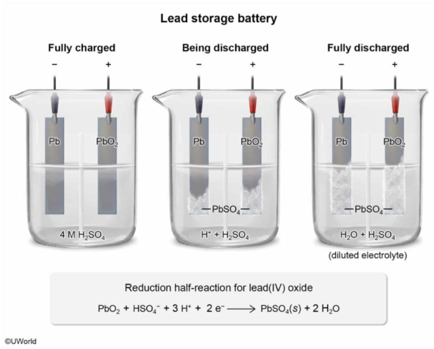
Reaction 4

Will the concentration of H_2SO_4 remain constant as a lead storage battery discharges?

- A. No, because HSO_4^- is consumed and H_2O is formed as the battery discharges. [62%]
- B. Yes, because $PbSO_4$ increases as the battery discharges. [8%]
- C. No, because the SO_4^{2-} ions are oxidized in the reaction. [16%]
- D. Yes, because the H^+ ions are not oxidized or reduced in the reaction. [13%]

Correct
62% answered correctly

Explanation:



Redox reactions occur when one or more electrons are transferred from one atom to another. The atom that **loses** electrons is **oxidized** (ie, electron donor) and the atom that **gains** electrons is **reduced** (ie, electron acceptor).

When a lead storage battery **discharges**, lead oxide (PbO_2) is reduced and reacts with aqueous HSO_4^- (generated from dissolved H_2SO_4) to form solid $PbSO_4$ and H_2O, according to Reaction 2. Initially, the concentration of H_2SO_4 is 4 M before discharging. As the battery discharges, HSO_4^- (formed from ionized H_2SO_4) is consumed and the leftover acid is further diluted by the water produced in the reaction. Therefore, the concentration of H_2SO_4 does not remain constant as the lead storage battery discharges.

(Choice B) Although the concentration of $PbSO_4$ increases as the battery discharges, this reaction requires SO_4^{2-} ions to precipitate out of solution, which *decreases* the concentration of H_2SO_4.

(Choice C) It is true that the concentration of H_2SO_4 does not remain constant; however, SO_4^{2-} is not one of the species being oxidized or reduced.

(Choice D) Although it is true that H^+ ions are neither oxidized nor reduced, the H_2SO_4 concentration does not remain constant because H^+ ions (from H_2SO_4) are used in the formation of water during the reduction half-reaction.

Educational objective:
A lead storage battery is an example of an oxidation-reduction (redox) reaction that occurs in an acidic electrolyte. Because the acid is consumed and water is one of the products of the redox reaction, the acidic electrolyte is diluted as the battery discharges.

Time spent: 0 sec
Subject:
General Chemistry

QID: 401283
System:
Solutions and Electrochemistry

An automated external defibrillator (AED) is a medical device used to send an electric shock to the heart after cardiac arrest. A key component of the AED is the power source, or battery. Batteries used in AEDs need to have a good charge-to-weight ratio; they must be safe and reliable as well as rechargeable.

Two types of rechargeable batteries used in early models of AEDs are the lead storage (also called lead-acid) battery and the nickel-cadmium (NiCd) battery. These batteries consist of multiple electrochemical cells that are connected in series to deliver an electric potential between 9 V and 18 V. A capacitor allows the AED to accumulate charge so that it can deliver between 300 V and 1,000 V.

Some AEDs use a sealed lead storage battery. Lead storage batteries are robust and hold a charge for a long time; however, they have a low energy-to-weight ratio. Each lead storage unit has a cell potential E_{cell} of approximately 2.0 V, and a battery of four cells weighing 1,000 g can provide 30 W·h of energy when discharged.

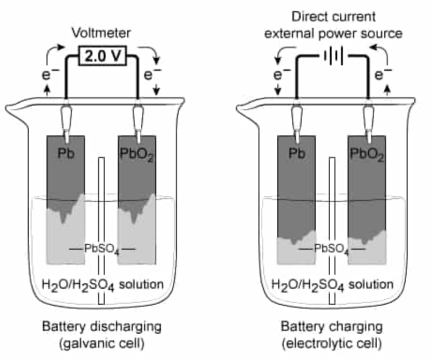


Figure 1 Electron flow in a lead storage battery when discharging and charging

The half-reactions for the anode and cathode of a discharging lead storage battery in 4 M sulfuric acid (H_2SO_4) are shown in Reactions 1 and 2, respectively:

$$\mathrm{Pb}(s) + \mathrm{HSO_4^-}(\mathrm{aq}) \rightarrow \mathrm{PbSO_4}(s) + \mathrm{H^+}(\mathrm{aq}) + 2\,\mathrm{e^-}$$

Reaction 1

$$\mathrm{PbO_2}(s) + \mathrm{HSO_4^-}(\mathrm{aq}) + 3\,\mathrm{H^+}(\mathrm{aq}) + 2\,\mathrm{e^-} \rightarrow \mathrm{PbSO_4}(s) + 2\,\mathrm{H_2O}(l)$$

Reaction 2

NiCd batteries have a greater energy-to-weight ratio than lead storage batteries but cannot hold as much charge. Each NiCd cell has an E_{cell} of approximately 1.3 V, and a single-cell battery weighing 120 g can provide 7.2 W·h of energy when discharged.

The half-reactions at the anode and cathode for a discharging NiCd battery in KOH(*aq*) are shown in Reactions 3 and 4, respectively:

$$\mathrm{Cd}(s) + 2\,\mathrm{OH^-}(\mathrm{aq}) \rightarrow \mathrm{Cd(OH)_2}(s) + 2\,e^-$$

Reaction 3

$$2\,\mathrm{NiO_2}(\mathrm{s}) + 2\,\mathrm{H_2O}(l) + 2\,\mathrm{e^-} \rightarrow 2\,\mathrm{Ni(OH)_2}(s) + 2\,\mathrm{OH^-}(\mathrm{aq})$$

Reaction 4

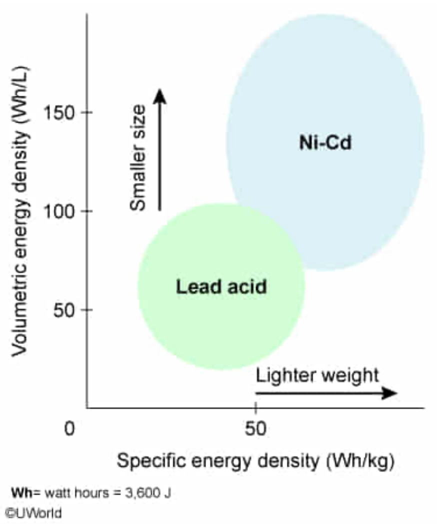
If energy density is the amount of energy per unit mass, which of the following statements must be true?

- ✔ A. NiCd batteries have a higher energy density than lead storage batteries. [82%]
- B. Lead storage batteries produce less energy than NiCd batteries. [5%]
- C. NiCd batteries produce more electrons per mole of metal than lead storage batteries. [9%]
- D. Both lead storage and NiCd batteries have the same energy density. [2%]

Correct

Correct
81% answered correctly

Explanation:



The amount of energy transferred by a battery is equivalent to the power (watts) supplied by the battery over an interval (eg, a watt-hour, W·h) and depends on the conversion of chemical energy into **electrical energy (joules)** by an oxidation-reduction reaction. **Energy density** is the ratio of the energy delivered by a battery to the battery's mass:

$$\text{Energy density} = \frac{\text{Energy}}{\text{Mass}} = \frac{\text{Power} \cdot \text{time}}{\text{Mass}}$$

As such, a battery with a high energy density has either a high energy output, a low weight, or both.

According to the passage, one of the advantages of NiCd batteries is that they have a higher energy-to-weight ratio than lead storage batteries. Even though lead storage batteries have a greater E_{cell} and energy output, NiCd batteries have more energy per unit mass and are smaller in size. Therefore, NiCd batteries have a higher energy density.

The difference in energy density between the batteries can be verified by calculating the values from the energy output and battery mass information given in the passage. A 1,000 g (1 kg) lead storage battery that provides 30 W·h of energy has an energy density of 30 W·h/kg:

$$\text{Energy density} = \frac{\text{Power} \cdot \text{time}}{\text{Mass}} = \frac{30 \text{ W} \cdot \text{h}}{1 \text{ kg}} = 30 \text{ W} \cdot \text{h/kg}$$

However, a 120 g (0.12 kg) NiCd battery that provides 7.2 W·h of energy has an energy density of 60 W·h/kg:

$$\text{Energy density} = \frac{\text{Power} \cdot \text{time}}{\text{Mass}} = \frac{7.2 \text{ W} \cdot \text{h}}{0.12 \text{ kg}} = 60 \text{ W} \cdot \text{h/kg}$$

(Choice B) Lead storage batteries have a cell potential of 2.0 V whereas NiCd batteries have a cell potential of 1.3 V. The amount of energy a battery can provide is directly proportional to its E_{cell}, so lead storage batteries produce *more* energy in absolute terms than NiCd batteries.

(Choice C) Based on the oxidation-reduction half-reactions for the NiCd battery and the lead storage battery, both reactions produce 2 moles of electrons per 1 mole of Cd or Pb that is oxidized.

(Choice D) Lead storage batteries have a lower energy-to-weight ratio compared to NiCd batteries, meaning lead storage batteries are heavier relative to the amount of energy they produce. Therefore, they do not have the same energy density.

Educational objective:
Energy density is the amount of energy produced by a battery per unit of mass. It is used to compare the energy output of batteries, or the work done by the battery.

Time spent:0 sec
Subject:
General Chemistry

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System:
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